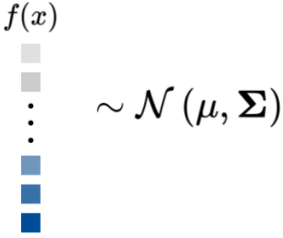
Introduction to Machine Learning

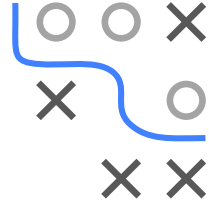
Gaussian Processes
Stochastic Processes and Distributions on Functions



Learning goals

- GPs model distributions over functions
- The marginalization property makes this distribution easily tractable
- GPs are fully specified by mean and covariance function
- GPs are indexed families

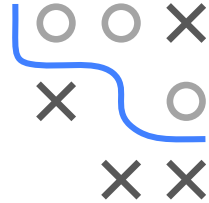
WEIGHT-SPACE VIEW



- Until now: hypothesis space $\mathcal{H}$ of parameterized functions $f(\mathbf{x} \mid \boldsymbol{\theta})$
- ERM: find risk-minimal parameters (weights) $\boldsymbol{\theta}$
- Bayesian paradigm: distribution over $\boldsymbol{\theta} \Rightarrow$ update prior to posterior belief after observing data acc to Bayes' rule

$$p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) = \frac{\text{likelihood} \cdot \text{prior}}{\text{marginal likelihood}} = \frac{p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \cdot q(\boldsymbol{\theta})}{p(\mathbf{y}|\mathbf{X})}$$

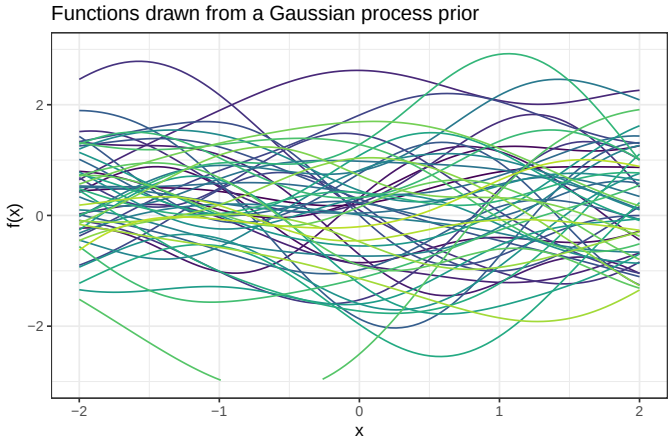
FUNCTION-SPACE VIEW



- New POV: rather than finding $\boldsymbol{\theta}$ to parameterize $f(\mathbf{x} \mid \boldsymbol{\theta})$ in parameter space, search space of admissible functions directly
- Sticking to Bayesian inference, specify prior distribution **over functions** and update according to observed data points

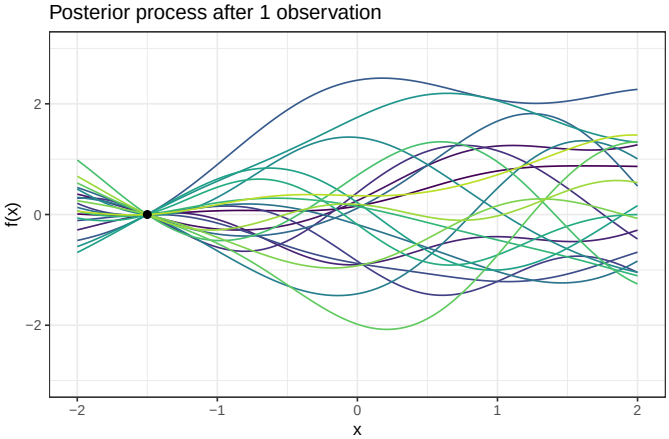
DRAWING FROM FUNCTION PRIORS

- Imagine we could draw functions from some prior distribution



DRAWING FROM FUNCTION PRIORS

- After observing some data points, we restrict sampling to functions consistent with the data



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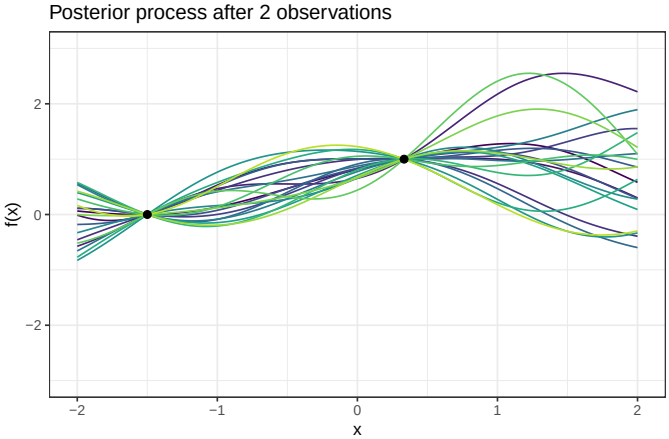

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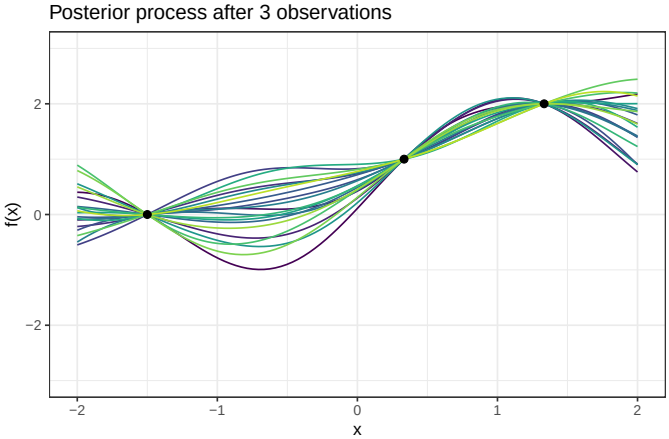
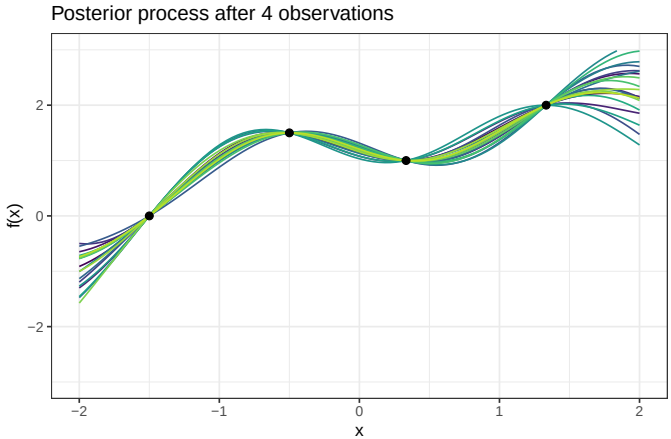


DRAWING FROM FUNCTION PRIORS

- Variety of admissible functions shrinks with observing more data



- Intuitively: distributions over functions have “mean” & “variance”

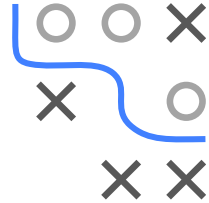
WEIGHT-SPACE VS. FUNCTION-SPACE VIEW

Weight-Space View

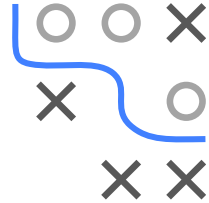
- Parameterize functions (e.g., $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$)
- Define distributions on $\boldsymbol{\theta}$
- Inference in param space Θ

Function-Space View

- Work on functions directly
- Define distributions on f
- Inference in fun space $\mathcal{H}$



DISCRETE FUNCTIONS



- Let $\mathcal{X} = \left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$ be a finite set
- Let $\mathcal{H} = \{h \mid h : \mathcal{X} \to \mathbb{R}\}$
- Any $h \in \mathcal{H}$ has finite domain with $n < \infty$ elements $\Rightarrow$ neat representation with n-dim vector

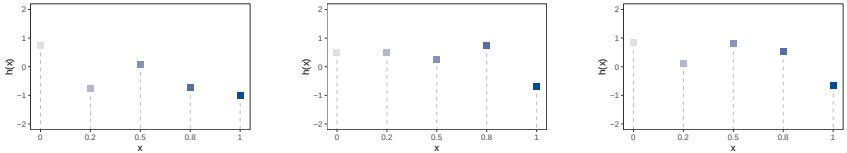
$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})]$$

EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Case 1: finite input space with cardinality $|\mathcal{X}| = 2$
- Example functions living in this space

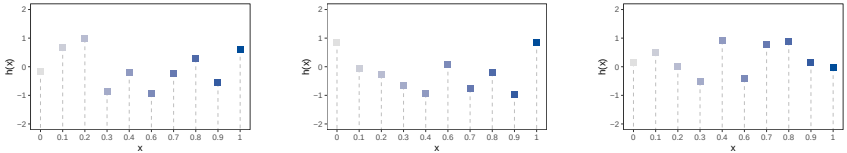
EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Case 2: finite input space with cardinality $|\mathcal{X}| = 5$
- Example functions living in this space

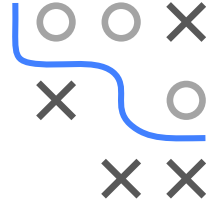


EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Case 3: finite input space with cardinality $|\mathcal{X}| = 10$
- Example functions living in this space



DISTRIBUTIONS ON DISCRETE FUNCTIONS



- Specify (discrete) probability function on functions $h \in \mathcal{H} \Rightarrow$ natural way: vector representation

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})]$$

- $\boldsymbol{h}$ as n-dim RV, e.g.,

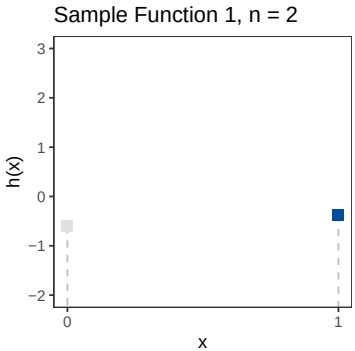
$$\boldsymbol{h} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

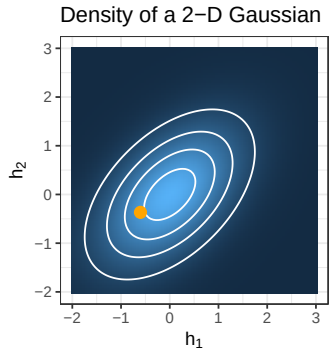
- For now: set $\mathbf{m} = \mathbf{0}$, assume $\mathbf{K}$ to be given

EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Case 1 (ctd): $h : \mathcal{X} \to \mathcal{Y}$ defined on 2 points
- Sample representatives by sampling from a 2-dim Gaussian

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), h(\mathbf{x}^{(2)})] \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$$





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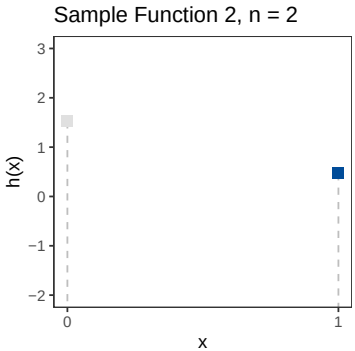

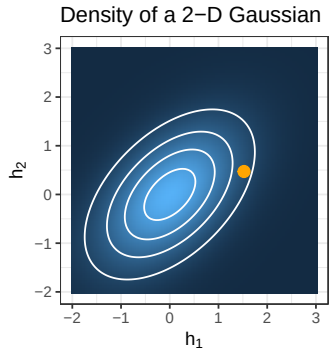

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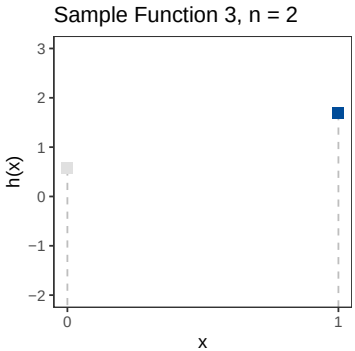

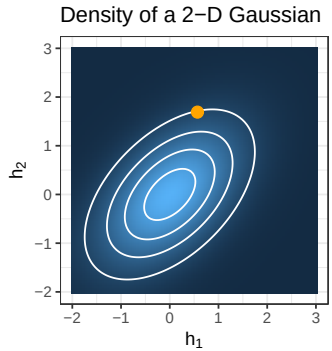

EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Case 2 (ctd): $h : \mathcal{X} \to \mathcal{Y}$ defined on 5 points
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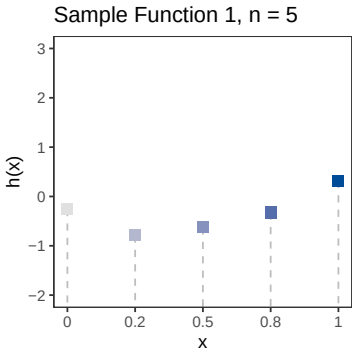
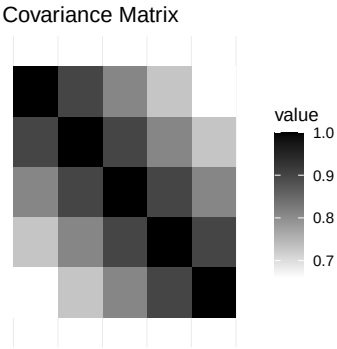
$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \dots, h(\mathbf{x}^{(5)})] \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$$





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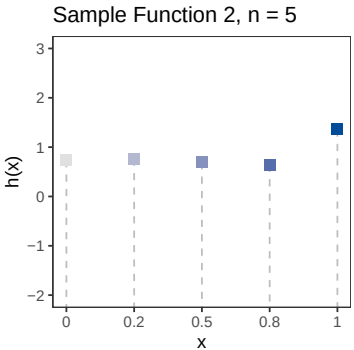

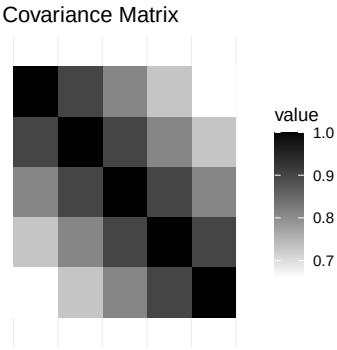

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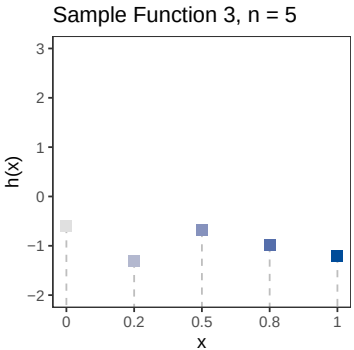

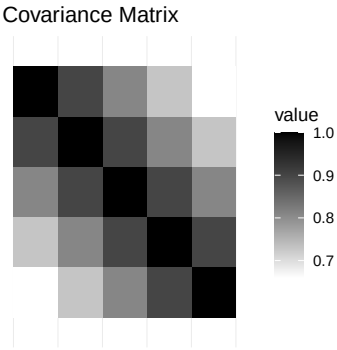

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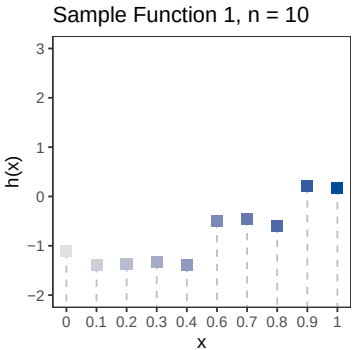

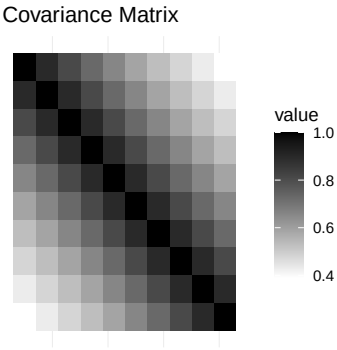

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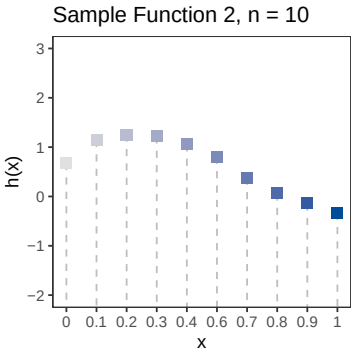

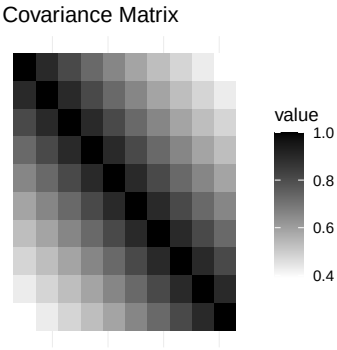

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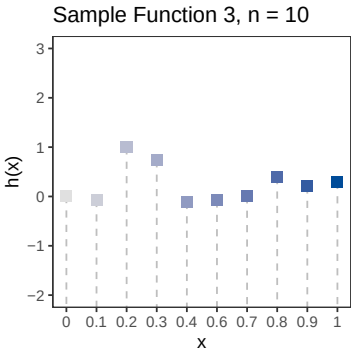
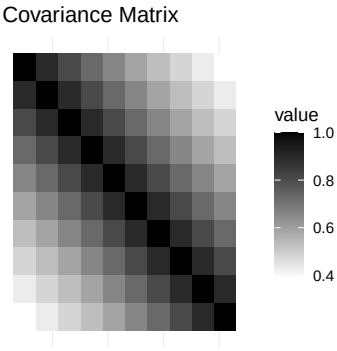


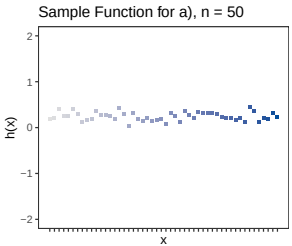


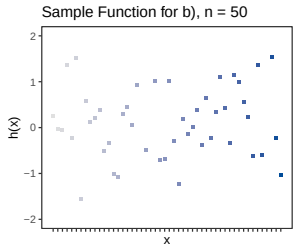
ROLE OF THE COVARIANCE FUNCTION

- Note: covariance controls “shape” of function drawn
- Extreme case a) almost perfectly correlated

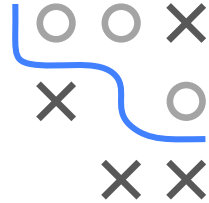
$$\mathbf{K} = \begin{pmatrix} 1 & 0.99 & \dots & 0.99 \\ 0.99 & 1 & \dots & 0.99 \\ 0.99 & 0.99 & \ddots & 0.99 \\ 0.99 & \dots & 0.99 & 1 \end{pmatrix}$$

- Extreme case b) uncorrelated $\Rightarrow \mathbf{K} = \boldsymbol{I}$





SPATIAL CORRELATION



- "Meaningful" functions (on numeric spaces) often characterized by a spatial property:

$$\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \text{ close in } \mathcal{X} \Rightarrow f(\mathbf{x}^{(i)}), f(\mathbf{x}^{(j)}) \text{ close in } \mathcal{Y}$$

- In other words: if points are close in $\mathcal{X}$, their function values should be correlated

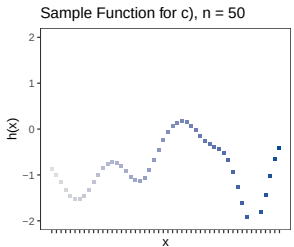
- Enforce this by choosing covariance function s.t.

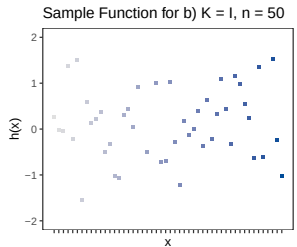
$$\mathbf{K}_{ij} \text{ high } \Leftrightarrow \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \text{ close in } \mathcal{X}$$

SPATIAL CORRELATION

- Compute covariance matrix by a function based on distance between $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$

- E.g., $\mathbf{K}_{ij} = k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \exp\left(-\frac{1}{2}\|\mathbf{x}^{(i)} - \mathbf{x}^{(j)}\|^2\right)$





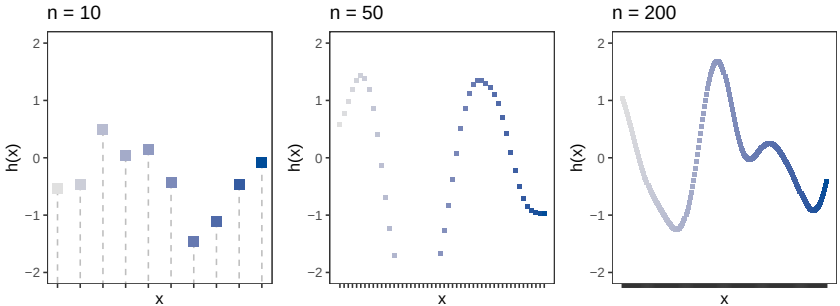
- More on covariance function, or **kernel**, $k(\cdot, \cdot)$ later on

FROM DISCRETE TO CONTINUOUS FUNCTIONS

- Recall: we defined distributions on functions by the (Gaussian) random vector of corresponding function values

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- We can do this for ever larger n (as "granular" as we want)



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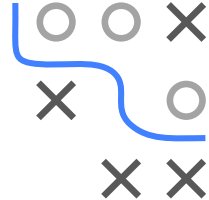
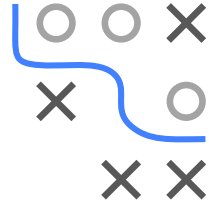


- No matter how large n is: still functions over discrete domains
- Can we simply extend our distribution def to **continuous**-domain functions by taking $n \to \infty$?
- Unclear how to obtain "infinitely" long (Gaussian) random vectors
- Observation: random vectors $\boldsymbol{h}$ are collections of RVs enumerated by index set $\{1, \ldots, n\} \Rightarrow$ **indexed family**
- Can we use more general, infinite index sets?

DEFINITION: INDEXED FAMILY



- Mathematical “rule” to map indices to some state space

$$s : T \to \mathcal{S}$$

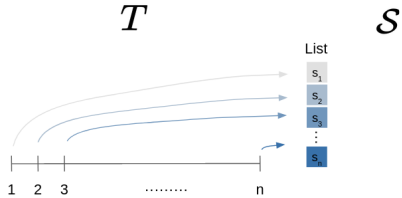
- Can assume many forms depending on T and $\mathcal{S}$

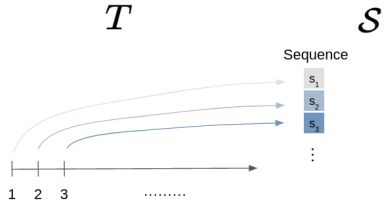
Cardinality of T	**$\mathcal{S}$: (real) numbers**	**$\mathcal{S}$: RVs**
Finite	list	random vector
Countably ∞	sequence	discrete-time **stochastic process**
Uncountably ∞	sequence	continuous-time **stochastic process**

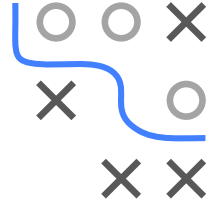
- $\mathcal{S}$ might also contain other objects like sets, . . .

EXAMPLE: REAL-VALUED INDEXED FAMILIES

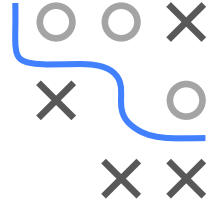
- $\mathcal{S} = \mathbb{R}$
- $t \mapsto s_t$
- Finite, e.g., $T = \{1, \ldots, n\}$
  $\Rightarrow$ list
- Countably ∞, e.g., $T = \mathbb{N}$
  $\Rightarrow$ sequence
- Uncountably ∞, e.g., $T = \mathbb{R} \Rightarrow$ example: $\{\sin(t) : t \in \mathbb{R}\}$







DEFINITION: STOCHASTIC PROCESS



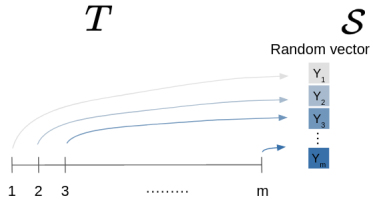
- Indexed family of RVs with specific covariance structure

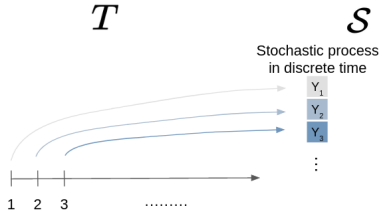
$$\{Y_t : t \in T\}$$

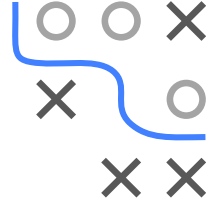
- Some distributional assumption for Y_t
- Often temporal interpretation of $T \Rightarrow$ e.g., stock prices
- Intuition: probability distributions describe random vectors, SP describe random functions

EXAMPLE: STOCHASTIC PROCESSES

- $\mathcal{S}$: space of RVs
- $t \mapsto Y_t$
- Finite, e.g., $T = \{1, \ldots, n\}$
  $\Rightarrow$ random vector
- Countably ∞, e.g., $T = \mathbb{N}$
  $\Rightarrow$ discrete-time SP
- Uncountably ∞, e.g., $T = \mathbb{R} \Rightarrow$ continuous-time SP

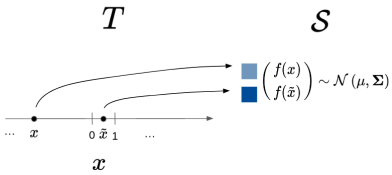


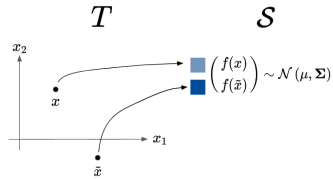




INTUITION: GAUSSIAN PROCESSES

- **Gaussian process**: special case of SP
- Index set: $\mathcal{X}$, cov structure: Gaussian
- Corollary: any finite random vector drawn from a GP is multivariate Gaussian $\Rightarrow$ **marginalization property**





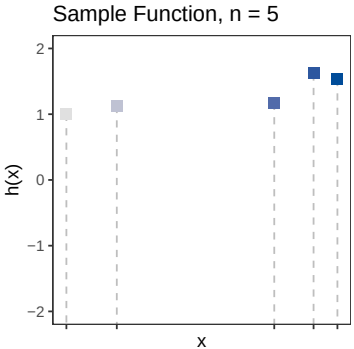
- Precisely what we were looking for: process to generate arbitrarily long Gaussian random vectors!

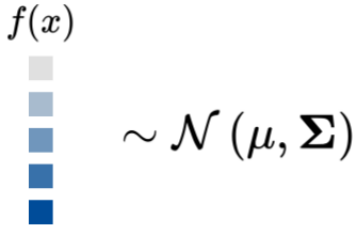
MARGINALIZATION PROPERTY

- Distinguishing feature of GPs: for any finite set of inputs,

$$\mathbf{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- **m** and **K** are calculated by a mean and cov function, resp





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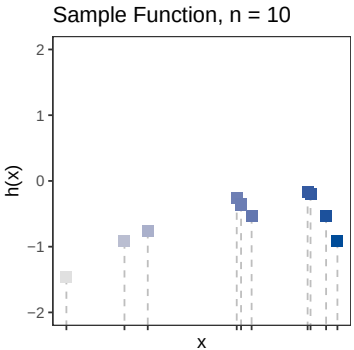

$$\sim \mathcal{N}\left(\mu, \mathbf{\Sigma}\right)$$

MARGINALIZATION PROPERTY

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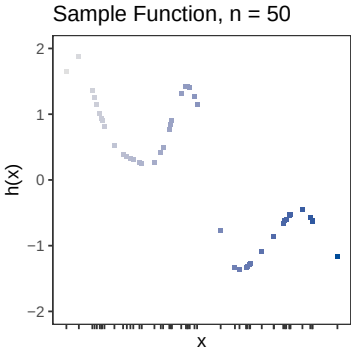
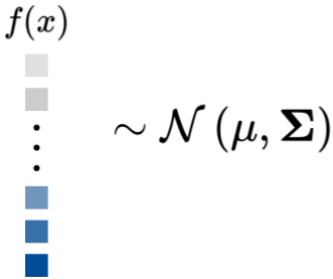
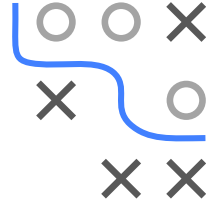




DEFINITION: GAUSSIAN PROCESS



- Formally,

$$f(\mathbf{x}) \sim \mathcal{GP}\left(m(\mathbf{x}), k\left(\mathbf{x}, \mathbf{x}'\right)\right)$$

iff for any finite set of inputs $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$,

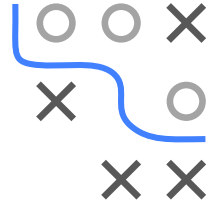
$$\mathbf{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

where

$$\mathbf{m} := \left(m(\mathbf{x}^{(i)})\right)_{i \in \{1,\ldots,n\}}, \quad \mathbf{K} := \left(k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})\right)_{i,j \in \{1,\ldots,n\}}$$

with **mean function** $m(\cdot)$ and **covariance function** $k(\cdot, \cdot)$

EXISTENCE THEOREM



- For **any**
  - state space $\mathcal{X}$,
  - mean function $m : \mathcal{X} \to \mathbb{R}$,
  - covariance function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$,

  there **exists** a Gaussian process $\mathcal{GP}(m(\cdot), k(\cdot, \cdot))$ s.t.

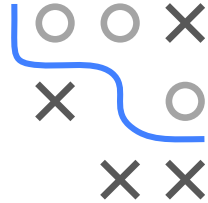
$$\begin{aligned} \mathbb{E}(f(\mathbf{x})) &= m(\mathbf{x}) \\ \mathrm{Cov}(f(\mathbf{x}), f(\mathbf{x}')) &= k\left(\mathbf{x}, \mathbf{x}'\right) \end{aligned}$$

  $\forall \mathbf{x}, \mathbf{x}' \in \mathcal{X}$

- Version of Kolmogorov consistency theorem
  $\Rightarrow$ proof: ▸ Grimmett and Stirzaker 2001 (Thm. 8.6.3)

- Intuition: for any mean & cov structure we prescribe for random vectors $\mathbf{f}$, there is a corresponding GP to generate them

DEFINITION: GAUSSIAN PROCESS



- GPs are thus completely specified by their mean & cov function

$$
\begin{aligned}
m(\mathbf{x}) &= \mathbb{E}[f(\mathbf{x})] \\
k(\mathbf{x}, \mathbf{x}') &= \mathrm{Cov}(f(\mathbf{x}), f(\mathbf{x}')) = \mathbb{E}\left[(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})])\,(f(\mathbf{x}') - \mathbb{E}[f(\mathbf{x}')])\right]
\end{aligned}
$$

- For now, we consider zero-mean GPs with $m(\mathbf{x}) \equiv \mathbf{0}$
  $\Rightarrow$ common, not necessarily drastic assumption

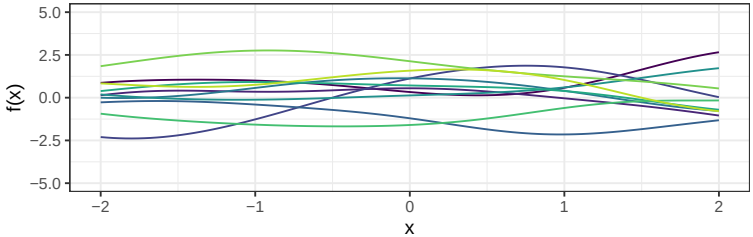
- Properties of GPs mainly governed by $k(\cdot, \cdot)$

- By virtue of existence thm: sampling from GPs gives us random functions with our properties of choice

SAMPLING FROM GAUSSIAN PROCESSES

- Example: $f(\mathbf{x}) \sim \mathcal{GP}(\mathbf{0}, k(\mathbf{x}, \mathbf{x}'))$ with cov function

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{x}'\|^2\right)$$

- To visualize sample functions,
  - choose high number n of (equidistant) points $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$
  - compute $\mathbf{K} = \left(k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})\right)_{i,j \in \{1,\ldots,n\}}$ from all pairs $\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$
  - draw $\boldsymbol{f} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{K})$

- 10 randomly drawn functions (note 0 mean)